

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 3 – Mock Interview

Course Code:	D6A63	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	C. Anu Keerthana	Reg. No.:	192472057
Section / Batch:		Date:	

Code of Conduct

I, C. Anu Keerthana (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: C. Anu Keerthana

Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatget voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse & Reference Resolution	Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text	Q1
Text Coherence & Discourse Structure	Analyzing coherence relations, discourse organization, and maintaining meaningful text flow	Q2
Dialog Acts & Interpretation	Identifying dialog acts, speaker intentions, and conversational meaning	Q3
Dialog and Conversational Agents	Architecture, components, and working of conversational agents/chatbots	Q4
Language Generation Architecture	Stages of Natural Language Generation (NLG), pipeline architecture, and applications	Q5
Surface Realization & Discourse Planning	Transforming semantic representations into grammatically correct sentences and planning discourse	Q6
Machine Translation – Transfer Approach	Translation process using transfer-based methods and practical applications	Q7
Machine Translation – Interlingua Approach	Language-independent representation and multilingual translation framework	Q8
Statistical Approaches in NLP	Statistical language models, probability-based language generation, and machine translation	Q9

"INTERVIEW QUESTIONS"

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NLP is a branch of AI that enables computers to understand, process, and generate human language. It combines linguistics, computer science, and ML.

The main goals are to understand human language, extract useful information, and generate meaningful responses. NLP aims to make human-computer communication.

Applications include chatbots, machine translation, sentiment analysis, speech recognition, and text summarization. Also used in search engines, spam detection.

NLP is the overall field of processing human language; NLU focuses on understanding its meaning. NLG focuses on generating human-like language from data or information.

Major challenges include ambiguity, sarcasm, context understanding, slang, multiple languages, and spelling variations. Understanding the same word in different contexts is particularly difficult.

Human language is complex, ambiguous, context-dependent,

and constantly changing. -the same sentence can
different meanings depending on context.

(7) Ambiguity occurs when a word, sentence, or phrase
more than one possible interpretation.

(8) Lexical Ambiguity occurs involves multiple word meanings.
Syntactic ambiguity involves multiple grammatical structures.
Semantic ambiguity involves multiple interpretations.

(9) A language model predicts the probability of words
or sequence of words in language.

(10) Rule-based NLP uses manually created linguistic rules and
Statistical NLP uses and learns patterns and probability
from large amounts of text data.

(11) A regular expression is a pattern used to search, match,
or manipulate text.

(12) Common symbols include *, +, ?, ., ^, \$, { } & ()
They represent repetition, optional characters, positions,
character sets and grouping.

mean zero or more occurrence of the preceding character.
 $a^* \Rightarrow "", a, aa, aaa, \dots$

+ mean one or more occurrences of the preceding character.
 $a^+ \Rightarrow a, aa, aaa, \dots$ (not empty sets)

* allows zero or more occurrence, while + requires at least one occurrence. $a^* \rightarrow$ match nothing but a^+ cannot.

Character classes specify a set or range of characters that can be matched. $[a-z]$ matches any lowercase letters.

FSA is a mathematical model that contains recognizer patterns, or strings using a finite set of number of states and it process input symbols and move between states.

18) An FSA contains state, an input alphabet, transition function, an initial state, and final states. Components determine how input strings are accepted or rejected.

19) FSAs are used for tokenization, morphological analysis, lexical processing and pattern matching. They can recognize valid word forms or linguistic patterns.

Happy → Unhappy is another example.

Morphological parsing breaks a word into its morphemes and identifies their grammatical roles.

It analyzes word structures using finite-state machines or transducers. It is useful for recognizing valid forms and identifying their morphological features.

Stemming reduces words to a common root or stem by removing prefixes or suffixes.

Lemmatization converts a word into its correct dictionary base called a lemma.

3) Stemming uses simple rules and may produce non-dictionary words, while lemmatization uses linguistic information.

34) Porter stemmer is an algorithm that removes common English suffixes to obtain word stems.

35) It can produce incorrect or non-meaningful stems because it mainly uses suffix-removal rules. It may also over-stem or under-stem words.

Unsmoothed N-gram model assigns probability directly from observed frequencies. Unseen N-grams receive probability zero.

If an N-gram never appears in training data, its probability becomes zero. This can make the probability of an entire sentence zero.

Smoothing assigns some probability to unseen N-grams by adjusting probability estimates. It helps language models handle words.

Smoothing prevents unseen N-grams from receiving zero probability. It improves the model's ability to handle new or rare sentences.

Backoff uses a lower order N-gram when a higher order N-gram has insufficient or zero evidence.

Deleted interpolation combines probabilities from different N-gram orders using weighted averages. (held-out or deleted training data).

Entropy measures the uncertainty or unpredictability of a language model. It is commonly expressed as the average information needed to predict the next word.

adjective describes or modifies a noun or pronoun.
beautiful, a beautiful flower → is an adjective.

adverb modifies a verb, adjective or another adverb.
quickly - 'she runs quickly' is an adverb.

determiner introduces or specifies a noun.
the, a, an, this and some

A preposition shows a relationship between a noun/pronoun and another word. ⇒ in, on, at, under and between.

A pos tagset is a predefined collection of labels used to represent grammatical categories.

A conjunction connects words, phrases or clauses.
⇒ and, but, or and because.

63) Pos ambiguity occurs when a word can belong to different grammatical categories.

- (64) Rule based Pos tagging assigns tags using manually designed linguistic rules and dictionaries. Rules use word forms and surrounding context.
- (65) Stochastic Pos tagging uses probabilities learned from a tagged corpus to select the most likely pos tag.
- (66) Transformation based tagging initially assigns tags and then applies correction rules based on context.
- (67) Rule-based tagging relies mainly on manually created linguistic rules. Stochastic tagging relies on statistical probabilities learned from training data.
- (68) A word can perform different grammatical functions depending on the sentence. $\Rightarrow$ light can be noun, adjective or verb.
- (69) In "I read a book" - book is a noun. "Book a flight." Book is a verb. The surrounding words and sentence context help determine the correct tag.
- 0) context helps determine the grammatical role of ambiguous words $\Rightarrow$ can is a modal verb, but also a noun.

CFG is a formal grammar used to describe the syntactic structures of sentences. It consists of rules that generate valid combinations of words.

(13) A CFG contains non-terminals, terminals, production rules, and a start symbol. These components define how valid sentences can be generated.

(14) A terminal is a basic symbol that appears directly in the generated sentences. "cat" & "runs" $\Rightarrow$ terminals.

(15) A non-terminal represents a grammatical category that can be expanded using production rules.

(16) A production rule specifies how a non-terminal can be replaced by other symbols.

(17) A parse tree is a hierarchical representation showing how a sentence is derived from grammar rules.

(18) A parse tree shows the complete grammatical derivation according to grammar rules. A syntax tree is often a simplified representation focusing on syntactic relationships.

(19) It means a sentence (S) can be formed using a noun phrase (NP) followed by a verb phrase. $S \rightarrow NP VP$

Agreement means the verb must agree with the subject in number and person as gender.

Subject-verb agreement means the verb must agree with the subject in number and person.

Subcategorization describes the types of complements that a word, especially a verb, can take.

A feature structure represents linguistic information using attributes and their values. A noun can have features such as number = singular & person = third.

A PCFG is a CFG in which each production rule has an associated probability. It helps choose the most likely parse when multiple parses are possible.

Discourse refers to connected language beyond individual sentences. It considers how sentences work together to communicate a complete message.

Discourse analysis studies relationships, meanings, structure and context across multiple sentences. It helps NLP systems understand how parts of a conversation are connected.

- (93) Reference resolution identifies which real-world entity or text element a referring expression points to.
- (94) Anaphora resolution identifies the earlier expression that a pronoun or phrase refers to.
- (95) Coherence is the logical and meaningful connection between ideas in a text. A coherence text has sentences that naturally relate to and support one another.
- (96) Discourse structure describes how difficult or different parts of text or conversation are organized and related.
- (97) Cohesion refers to linguistic connections between sentences such as pronouns and conjunctions and it refers to the overall logical meaning.
- (98) A dialogue system is an NLP application that communicates with users through natural language.
- (99) A conversational agent is an AI system designed to interact with users through text or speech. It understands user inputs and generates responses.
- (100) A dialogue act represents the function or intention of an utterance in a conversation. Ex- includes question, answer etc.

l-world entity
points to.

version that

Case Study (Analytical & Problem Solving)

Annexure A – Mock Interview
Rubrics for Calculation

Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation

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Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	20	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1	25	25	20	10	10	90	90
2	20	20	20	15	15	90	
3	25	25	20	10	10	90	
4	20	20	20	15	15	90	
5	25	25	20	10	09	89	
6	20	20	20	15	15	90	
7	25	25	15	15	15	90	
8	25	25	20	10	10	90	
9	20	20	20	15	15	90	
10	25	25	20	10	10	90	

Faculty In-charge	Course Coordinator	Program Director